



**STRUCTURAL DESIGN AND FEASIBILITY ASSESSMENT OF
LOCALLY SOURCED ALUMINUM AND COMPOSITE MATERIALS FOR
A CUBESAT STRUCTURAL ENGINEERING MODEL**

**A Project Report Submitted to the Department of Aerospace Engineering,
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in Aerospace Engineering (First Semester Defense).**

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PHASE III: Factor of Safety & Structural Validation Report

Material Scope: Aluminum 6063-T6 (Frame) and CF-Nylon / PA-12 (Bulkheads)

1.0 Quasi-Static Strength Validation (15g Axial Load)

Validation Criterion: Factor of Safety (FoS) > 2.0 under 15g loading.

Formula: $\text{FoS} = \text{A-basis Yield Strength} / \text{Maximum von Mises Stress}$

1.1 Aluminum 6063-T6 Frame

A-basis Yield Strength: 162.8 MPa

Maximum Applied Equivalent Stress (Static): 0.78 MPa

Calculated FoS: $162.8 / 0.78 = 208.7$

Status: PASS

The static FoS significantly exceeds the 2.0 minimum requirement.

1.2 CF-Nylon 12 (PA12) Panels

Z-Axis Yield Strength: 6.6 MPa

Maximum Applied Z-Axis Normal Stress (Static): 0.046 MPa

Calculated FoS: $6.6 / 0.046 = 143.4$

Status: PASS

The panels are highly stable under static inertial launch loads.

2.0 Dynamic Strength Validation (Random Vibration)

Validation Criterion: 3-Sigma dynamic stresses must remain below specific percentage thresholds of the material yield strength to prevent high-cycle fatigue and dynamic yielding.

2.1 Aluminum 6063-T6 Frame

Criterion: 3σ von Mises / Yield < 70%

A-basis Yield Strength: 162.8 MPa

Maximum 3σ Equivalent Stress: 71.726 MPa

Stress Ratio: $(71.726 / 162.8) * 100 = 44.05\%$

Status: PASS

The dynamic stress uses only 44.05% of the allowable yield, safely satisfying the <70% limit.

2.2 CF-Nylon 12 (PA12) Panels (Out-of-Plane Z-Axis)

Criterion: 3σ Z-stress / Z-UTS < 50% (Evaluated using Yield of 6.6 MPa)

Out-of-Plane (Z-Axis) Yield Strength: 6.6 MPa

Maximum 3σ Normal Stress (Z-Axis): 3.1306 MPa

Stress Ratio: $(3.1306 / 6.6) * 100 = 47.43\%$

Status: PASS

The geometric modifications successfully reduced the peak dynamic stress. The stress ratio of 47.43% now falls safely below the strict 50% threshold defined in Table 3.4.2.

3.0 Summary Conclusion

The CubeSat assembly has fully cleared Phase III structural simulations. The Aluminum 6063-T6 primary structure passes all criteria with massive margins, validating its flight readiness. Furthermore, the CF-Nylon 12 panels have achieved a 47.43% dynamic stress ratio in the critical Z-axis, successfully passing the conservative 50% limit and eliminating the risk of out-of-plane delamination. The engineering model is mathematically validated and cleared to proceed to Phase IV physical fabrication.